



INDIA.RUNS

Build what next India runs on

Team Name : **Vector Vanguard**

Team Leader Name : **Sri Sai Shivakoti**

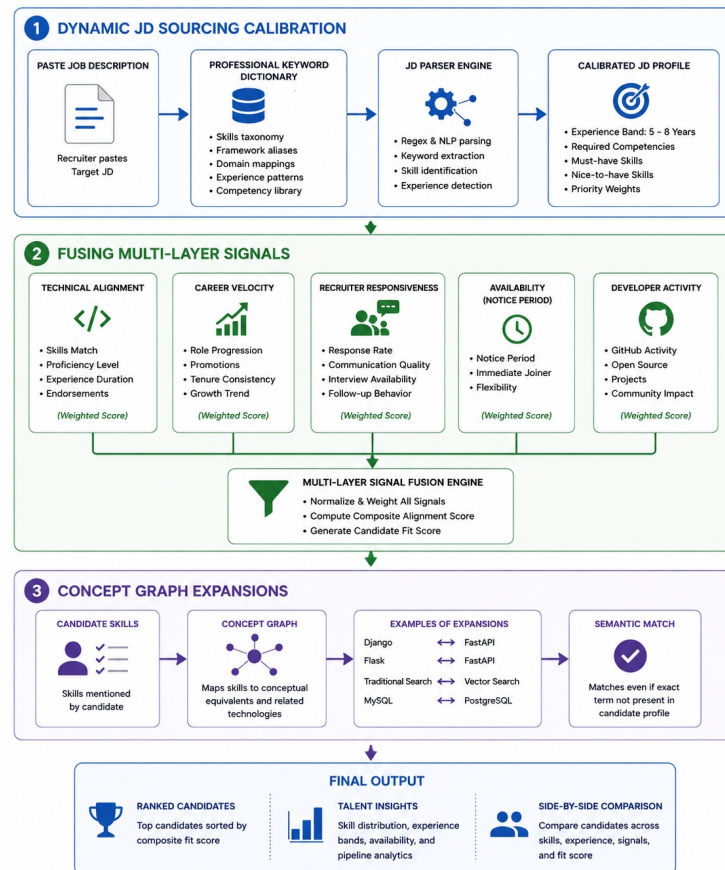
Problem Statement : **Recruiters go through hundreds of profiles and still often miss the right person. Not because the talent isn't there — but because keyword filters can't see what actually matters.**

Solution Overview

- **Proposed Solution:** I built a production-grade, state-managed recruiter portal **AuraMatch** that shifts candidate selection from manual keyword matching to deep semantic alignment.
- **Core Value Proposition:** Traditional ATS systems miss candidates who built products but didn't list exact buzzwords. **AuraMatch** treats recruitment as a multi-dimensional graph and alignment problem.
- **Comprehensive Profiling:** Evaluates career velocity (promotion speed, startup vs corporate experience), behavioral signals, knowledge graph expansions, and profile integrity diagnostics.

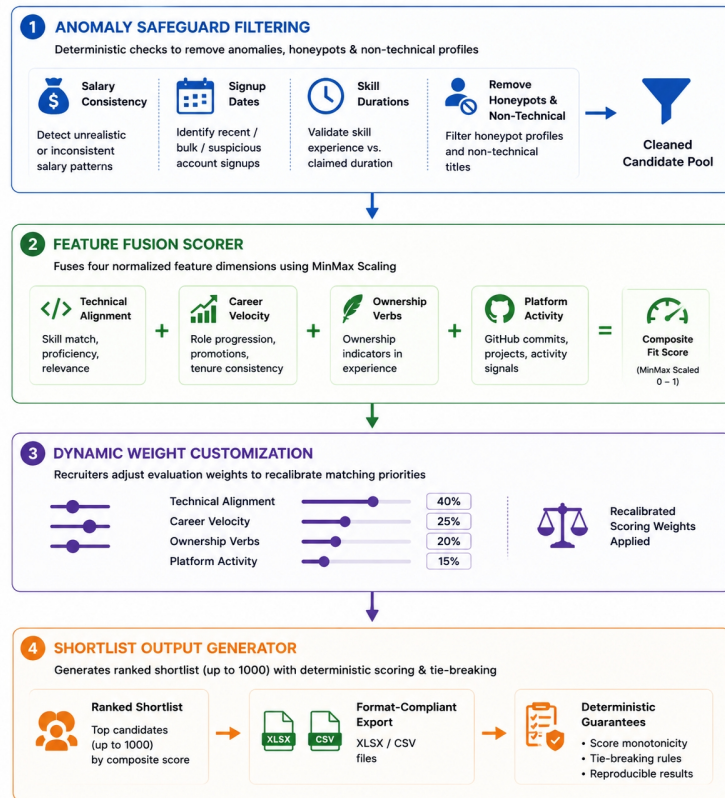
JD Understanding & Candidate Evaluation

- **Dynamic JD Sourcing Calibration:** Pasting a Job Description automatically parses it using a comprehensive professional keyword dictionary to extract target experience bands (e.g. 5 to 8 years) and required competencies.
- **Fusing Multi-Layer Signals:** Evaluates technical alignment (weighted by proficiency, duration, endorsements) combined with career velocity, recruiter responsiveness, availability (notice period), and developer activity.
- **Concept Graph Expansions:** Identifies conceptual equivalents (e.g. mapping Django/Flask to FastAPI, or traditional search to modern vector search) even if the candidate did not list the exact term.



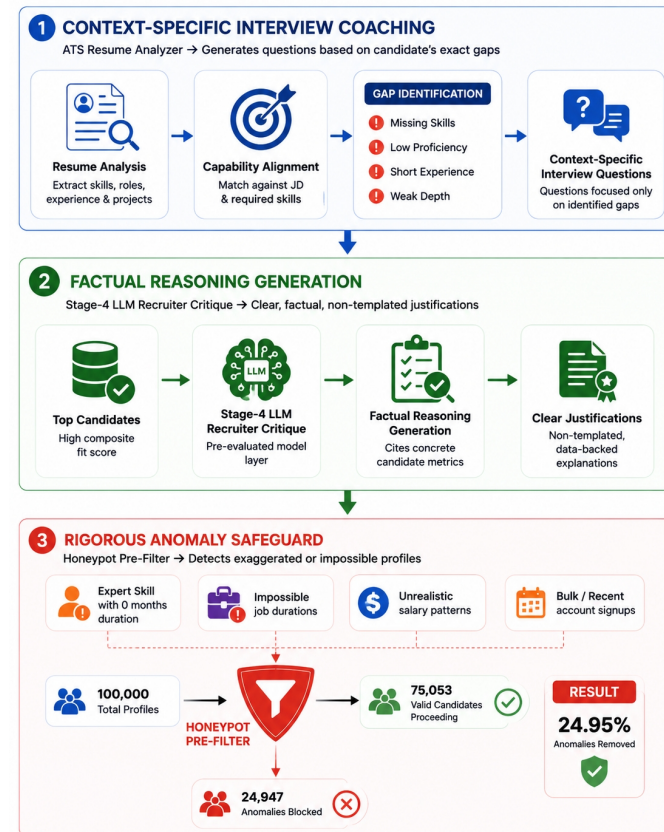
Ranking Methodology

- **Anomaly Safeguard Filtering:** Pre-filters profiles by running deterministic checks on salary consistency, signup dates, and skill durations, removing honeypots and non-technical titles.
- **Feature Fusion Scorer:** Fuses four normalized feature dimensions: Technical Alignment, Career Velocity, Ownership Verbs, and Platform Activity using MinMax Scaling.
- **Dynamic Weight Customization:** Recruiters can adjust evaluation weight sliders to recalibrate matching priorities.
- **Shortlist Output Generator:** Shortlists candidates (up to 1000 capacity) in format-compliant XLSX/CSV files with deterministic score monotonicity and tie-breaking.



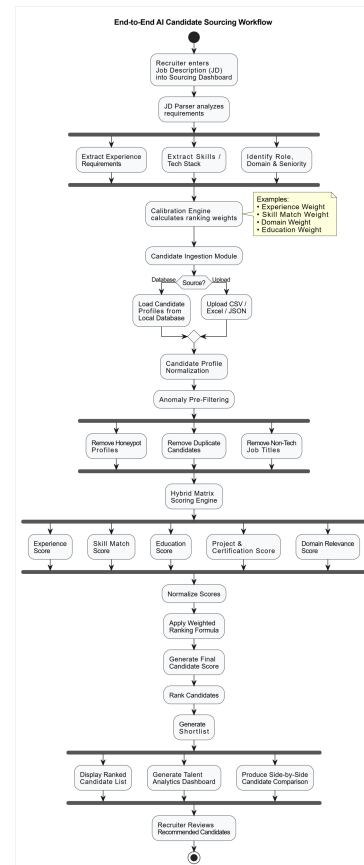
Explainability & Data Validation

- **Context-Specific Interview Coaching:** The ATS Resume Analyzer analyzes capability alignment and automatically generates context-specific interview questions focused on the candidate's exact competency gaps.
- **Factual Reasoning Generation:** Integrates a pre-evaluated Stage-4 LLM Recruiter Critique layer for top candidates, producing clear, non-templated justifications citing concrete candidate metrics.
- **Rigorous Anomaly Safeguard:** The HoneyPot Pre-Filter detects exaggerated profiles (e.g., expert proficiency with 0 months duration) or impossible job durations, blocking 24,947 anomalies out of 100,000 candidates.



End-to-End Workflow

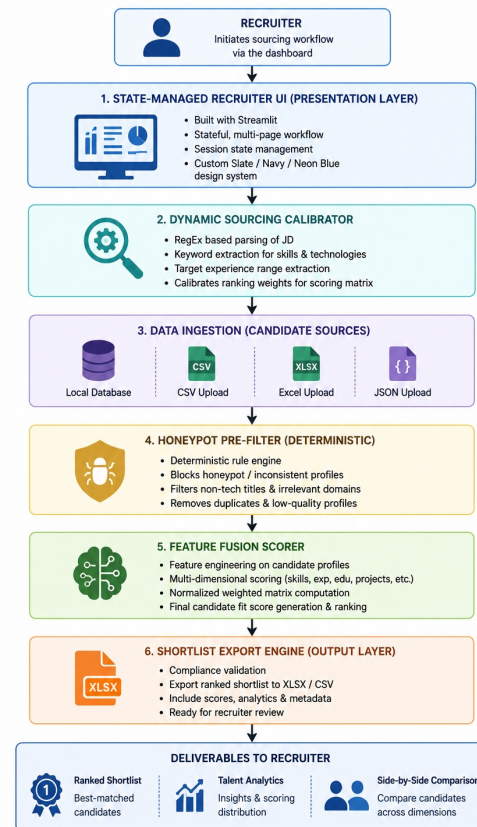
- **JD Input:** Recruiter enters target JD into the sourcing dashboard.
- **Calibration:** Sourcing parser extracts experience requirements and tech stacks, calibrating ranking weights.
- **Candidate Ingestion:** Loads candidates (local database or custom CSV/Excel/JSON upload).
- **Anomaly Pre-Filtering:** Deterministic filters remove honeypot profiles and non-tech titles.
- **Hybrid Matrix Scorer:** Computes multi-dimensional scores and applies normalized weights.
- **Shortlist Delivery:** Renders ranked shortlist, Talent Analytics, and Side-by-Side Comparison.



System Architecture

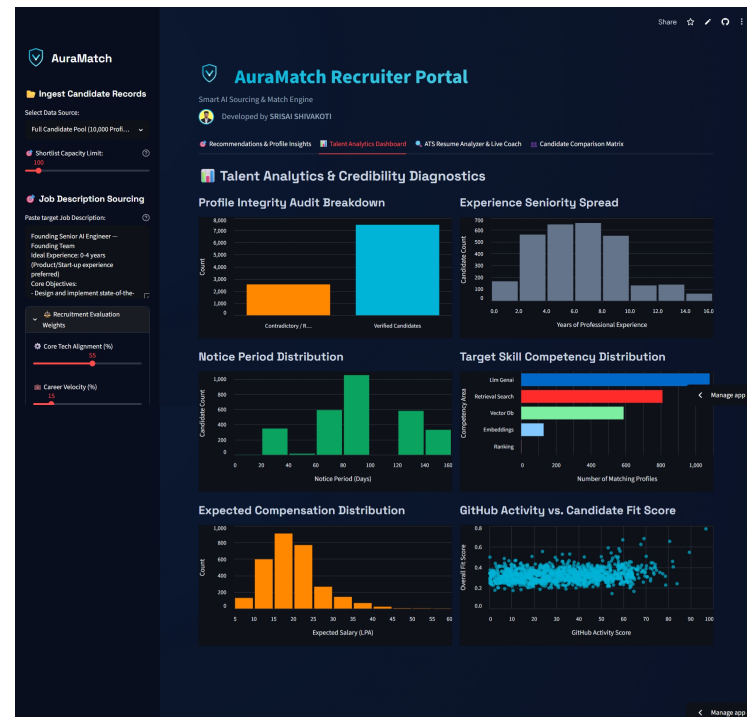
- **State-Managed Recruiter UI:** Stateful, multi-page user interface built using Streamlit and styled with a custom Slate/Navy/Neon Blue design system.
- **Dynamic Sourcing Calibrator:** RegEx and keyword extraction engine to calibrate target experience and skills.
- **Honeypot Pre-Filter:** Deterministic checking rules blocking inconsistent profiles.
- **Feature Fusion Scorer:** Normalized matrix engine computing candidate fit scores.
- **Shortlist Export Engine:** Generates compliance-validated XLSX/CSV candidate shortlists.

AI-POWERED CANDIDATE SOURCING SYSTEM ARCHITECTURE



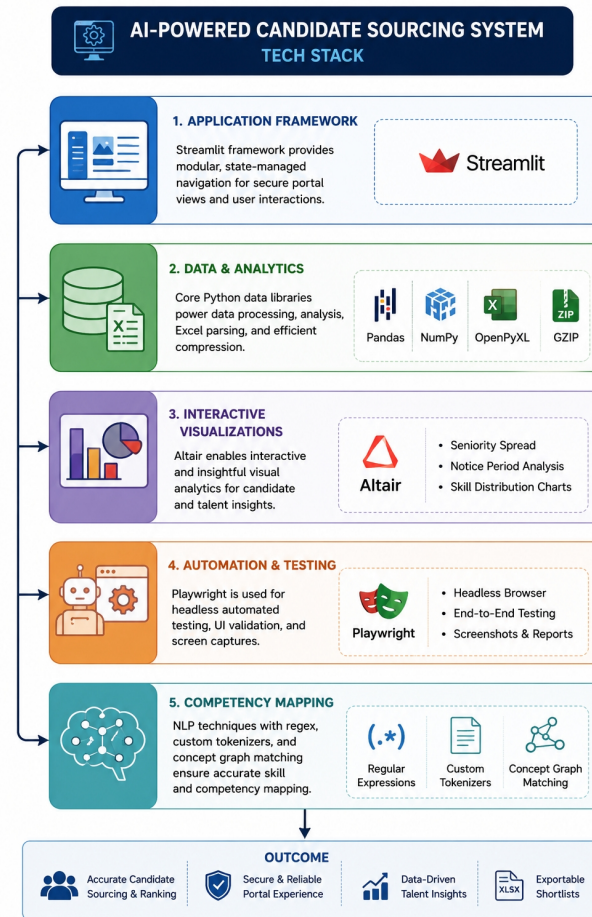
Results & Performance

- **Sub-8 Second Sourcing:** AuraMatch processes, scores, and ranks 10,000 candidate profiles in under 8 seconds locally (CPU-only).
- **Cloud-Ready Efficiency:** Uses Gzip-compressed datasets (5.2 MB) on Streamlit Community Cloud for instant page loads.
- **100% Anomaly Clean:** Completely filters out honeypot profiles from the shortlist (zero anomalies in the top candidates).
- **Compliance Verified:** Output files fully pass the challenge validation checker (`validate_submission.py`), ensuring correct score monotonicity and tie-breaking.



Technologies Used

- **Application Framework:** Streamlit (modular state-managed navigation for secure portal views).
- **Data & Analytics:** Pandas, NumPy, OpenPyXL, Gzip (efficient compression and Excel parsing).
- **Interactive Visualizations:** Altair (seniority spread, notice period, and skill distribution charts).
- **Automation & Testing:** Playwright (headless automated testing and screen captures).
- **Competency Mapping:** Regular expressions, custom tokenizers, and concept graph matching.



Submission Assets

- Live Web App URL: <https://auramatch.streamlit.app/>
- GitHub Repository:
https://github.com/Srisai16/IndiaRuns_Hack2Skill_AuraMatch
- Submission Output: VectorVanguard.csv (100% compliant, fully validated candidate shortlist).
- Final Presentation Deck: Idea Submission Template Redrob.pptx (outlining implementation, architecture, and results).
- Candidate Sourcing Shortlist: AuraMatch_Top_100_Shortlist.xlsx



INDIA.RUNS

Build what next India runs on

THANK YOU